

# ADRIAN MATIAS

aadrian.mat@gmail.com | +34 654 435 598 | linkedin.com/in/adrianmatiasdiez | github.com/adrianmatias | x.com/adrian2mat |  
adrianmatias.github.io *Calle Miguel Cordero del Campillo 9, 24007 León – Spain*

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## PROFILE

Data Scientist / ML Engineer with 12 years shipping end-to-end machine learning from problem framing to production. Deep roots in core DS — ad-tech ranking & user acquisition (Appodeal, Chartboost), fraud detection, NLP (RavenPack), recommender systems, Bayesian Bandits / RL, and large-scale experimentation on Spark / Databricks — with current work in GenAI: VET, PwC Advisory's LLM-driven vendor-evaluation platform (Azure OpenAI via Cavendish) — grounding raw vendor documentation into cited evidence, scoring engines, and contract-first pipelines. Held the seat the team labeled backend architect & correctness gatekeeper. Approach driven by simplicity, first-principles thinking, and measurable business outcomes.

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## EXPERIENCE

### Senior AI Engineer

**Cavendish Professionals → PwC Advisory (long-term contract)** | León, Spain – Remote | *May 2026 – Present*

Senior AI Engineer on VET (Vendor Evaluation Tool), an LLM-driven platform PwC Advisory uses to evaluate vendors with contractually binding outcomes: ingest heterogeneous vendor submissions and raw documentation → ground extraction in cited evidence → score → compare. Ten weeks, four seats held in sequence — backend feature developer, contract & schema owner, assessment-workbook lane owner — finishing by deciding what counts as correct. The last seat carries the most weight per hour and the least visible output: when a budget review counts merged features, it also carries the least protection. Stack: Python, FastAPI, Pydantic, PostgreSQL, Azure OpenAI, Next.js, GitHub Actions; deployed on internal AKS.

#### Where I sat — the team's label, backed by the review record

- **Backend architect & correctness gatekeeper** — the label the team's own page gave the seat; the review record supports it without any narrative help.
- **A pair, not a hierarchy** — review traffic with the senior counterpart ran symmetric, 212 against 204; every hard call about transport, tenancy, provenance and lane separation settled between those two seats — and it settled toward less coupling.
- **Carried correctness alone where needed** — 46 of 139 pull requests reached a decision with no second pair of eyes: the honest edge of being the gate.

#### The strongest thing

- **Made correctness executable instead of stating it** — enforcement artifacts each move a belief out of one head into a place that fails without its author: the CI gate set, the machine-readable requirement contract tree, the schema-completeness gates, and formal request-change verdicts. That skill transfers to any team, and it outranks the merged-PR count.
- **Built for the inheritor** — the enforcement landed in the final two weeks of the ten-week engagement, so most of its value accrues to whoever inherits the repository — by design, not by accident.

#### What VET is

- AI-assisted vendor evaluation for contractual, binding decisions: vendor submissions and documentation grounded into cited, auditable evidence per requirement; heterogeneous data aggregated by domain experts into decision-ready scores.
- Built on Azure OpenAI with a contract-first backend (typed Pydantic contracts, CI enforcement, reproducible local assessment runs). Not yet in production — the engagement was measured on delivered correctness, not live traffic.

### Data Scientist → DS Technical Lead

**Appodeal:** appodeal.com | León, Spain – Remote | *Dec 2024 – Jan 2026*

Developed install prediction models for mobile user acquisition: cold-start, transformers, user embeddings, transfer learning, MLOps, production pipelines, parity resolution, experimentation, and monitoring. Progressed toward technical leadership based on ownership and strategic perspective.

#### Key Achievements

- Unified modeling across targets, advertisers, and platforms via attention/transformer architecture, increasing end KPI.
- Enabled seamless advertiser onboarding through novel cold-start techniques (custom PyTorch + transfer learning).
- Resolved parity mismatches and calibration gaps; productionized parity workflows with unit tests and decoupling in Databricks.
- Transfer-learning user embeddings achieving performance parity with orders-of-magnitude cost savings.

**Senior Data Scientist**

**Quant AI Lab:** quant.global | León, Spain – Remote | *Jul 2024 – Dec 2024*

Designed and delivered end-to-end ML solutions for international clients; reviewed technical scope, aligned with stakeholders, and supervised model productivization.

**Key Contributions**

- Led rapid turnaround of underperforming client ML system (utility fraud detection & ranking): diagnosed architectural issues, executed a surgical plan, restored accuracy/efficiency, and recovered client satisfaction.
- Discovered critical issues in data ingestion and feature-engineering pipelines.
- Led real-time credit card fraud classifier combining scalable ML with deep VAE. Open version: [github.com/adrianmatias/ml-sandbox/tree/master/tx\\_class](https://github.com/adrianmatias/ml-sandbox/tree/master/tx_class)

**Senior Data Scientist**

**Chartboost:** chartboost.com | León, Spain – Remote | *Jun 2023 – Jun 2024*

Developed ad ranking and reinforcement learning systems for in-app game advertising; advanced DSP performance through large-scale experimentation and behavioral analysis.

**Key Achievements**

- Evolved and monitored Bayesian Bandit RL system with policy evaluation tools; designed tailored online/offline evaluation frameworks.
- Built Two-Towers neural ranking algorithm based on recommender systems.
- Post-mortem analysis of adversarial DSP dynamics and model feedback loops.

**Machine Learning Engineer**

**RavenPack** | León, Spain – Remote | *Mar 2022 – Jan 2023*

Researched, implemented, and maintained text analytics, NLU, and semantic search products on massive financial news data; productionized prototypes with emphasis on MLOps and reproducibility.

**Key Achievements**

- Few-shot search personalization: simplified architecture, reduced error 30%.
- Active learning PoC for learn-to-rank: projected 300% annotation efficiency gain.
- Novel long-text fingerprinting algorithm for duplicate content detection.

**Earlier Roles**

**Numbrs** *Aug 2020 – Oct 2021*

Extended Deep Learning / Active Learning for transaction processing; scaled classifier across countries with GDPR-compliant pipelines.

**Stratio BD** *Aug 2019 – Aug 2020*

Propensity/churn on Spark; BERT+graph email segmentation; multi-source weather DB (Elasticsearch/Flask).

**Diligant** *Jun 2015 – Apr 2017*

Redesigned Consumer Persona ML model; Game Theory attribution prototype; co-directed Master's thesis.

**IEBS School** *Oct 2019 – Oct 2021*

Webinars and masterclasses on real-time Big Data, AI, and ML applications.

**DOCOMO DIGITAL** *Apr 2017 – Aug 2019*

Led fraud detection (99.8% specificity, <50 ms) and default-payment filtering; mentored juniors.

**2007–2015**

Civil/tunnel engineering; university teaching: Math, Physics, Chemistry.

**Freelance**

**Freelance Data Scientist / ML Engineer** | *Mar 2022 – Jun 2025*

On-demand projects in semantic search, time series forecasting, and portfolio optimization for financial analysis.

**SKILLS**

| Core Technologies                                                                                 | Machine Learning & GenAI                                                                                                       |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Python, FastAPI, Pydantic, PostgreSQL, PyTorch, Transformers, Azure OpenAI / Azure / AKS, AWS/S3, | LLMs, RAG, agentic systems, LangChain / LangGraph-style orchestration, document extraction & provenance, scoring engines, Deep |

## Core Technologies

Databricks, PySpark, Pandas, SQL, Docker, MLflow, Git, GitHub Actions, AI-augmented tools (Ollama, llama.cpp, Hermes Agent)

## Engineering Practices

Clean code, typed contracts as source of truth, CI drift gates, OOP & FP, TDD / local e2e harnesses, reproducible pipelines, Agile / XP, production ownership

## Machine Learning & GenAI

Learning (attention, embeddings, transfer learning), Recommender Systems, NLP, RL / Bayesian Bandits, MLOps (parity, A/B, monitoring), Few-Shot / Active Learning, Ranking

## Consulting & Soft Skills

Multi-client consulting delivery, stakeholder communication, correctness gatekeeping & requirements arbitration, mentoring, business impact focus, simplification / unification

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## SELECTED COURSES & CERTIFICATIONS

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|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| - Claude Code: A Highly Agentic Coding Assistant: Coursera (2026)       | - Functional Programming in Scala Specialization: Coursera (9.2/10, 2017)              |
| - ML Engineering for Production (MLOps) Specialization: Coursera (2021) | - Bitcoin and Cryptocurrency Technologies: Coursera (9.5/10, 2017)                     |
| - Deep Learning with PyTorch: Siamese Networks (2022)                   | - Algorithms & Data Structures with Java (I, II): Coursera (8.5/10, 2014)              |
| - Deep Learning Specialization: Coursera (2016)                         | - Analytics Edge: Data Science & Predictive Models: MITx (9.3/10, 2014)                |
| - Structuring Machine Learning Projects: Coursera (2018)                | - Introduction to Programming & Data Science Using Python (I, II): MITx (8.7/10, 2014) |
| - Apache Spark for Big Data & Scalable ML: Coursera (9.9/10, 2015)      |                                                                                        |

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## SELECTED PROJECTS

- **not the tao with adrian matias:** [youtube.com/@notTheTaoWithAdrianMatias](https://youtube.com/@notTheTaoWithAdrianMatias)  
Podcast exploring simplicity, consciousness, reality, machine learning, coding, sovereign AI.
- **RAG Blog System:** [github.com/adrianmatias/ml-sandbox/tree/master/ragblog](https://github.com/adrianmatias/ml-sandbox/tree/master/ragblog)  
Retrieval-augmented generation pipeline integrating LLMs, vector embeddings, and web crawling.
- **The Mirror:** <https://github.com/adrianmatias/the-mirror>  
Knowledge repository based on llm-wiki for personalized assistant and insight for agentic AI.
- **Credit Card Transaction Fraud Classifier:** [github.com/adrianmatias/ml-sandbox/tree/master/tx\\_class](https://github.com/adrianmatias/ml-sandbox/tree/master/tx_class)  
Real-time fraud detection API (classical + unsupervised signals). Deploy-ready: FastAPI, Docker, RayServe.

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## EDUCATION

- **Master's in Geological Engineering:** University of Oviedo (2010–2012)
- **Degree in Mining & Energy Engineering:** University of León (2012)
- **Bachelor in Mining Engineering:** University of León (2003–2007)

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*References available on request*